Twelve-fold way Counting

\# of maps $\varphi: A \to B$, $|A| = n$, $|B| = m$

A: $\{ \circ \ \cdots \ \circ \}$ B: $\{ \square, \square, \cdots \square \}$

different — d undistinguishable — u

A	B	φ	#
d	d	arb	m^n
d	d	inj	$(m)_n = m(m-1)\cdots(m-n+1)$
d	d	sur	$m!\ S(n,m)$
d	u	arb	$\sum_{k=1}^{m} S(n,k)$
d	u	inj	$\delta(n \le m)$
d	u	sur	$S(n,m)$
u	d	arb	$\binom{n+m-1}{m-1} = \binom{n+m-1}{n}$
u	d	inj	$\binom{m}{n}$
u	d	sur	$\binom{n-1}{m-1}$
u	u	arb	$\sum_{k=1}^{m} P(n,k)$
u	u	inj	$\delta(n \le m)$
u	u	sur	$P(n,m)$

$$C(n,k) = \binom{n}{k} = \frac{n!}{k!\,(n-k)!}$$

<u>Thm</u> $(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}$

Thm: (Vandermonde's identity) $\binom{m+n}{k} = \sum_i \binom{m}{i}\binom{n}{k-i}$

Thm: (Chu's indentity) $\binom{m+n+1}{n+1} = \sum_{k=0}^{m} \binom{n+k}{n}$

Pf: $LHS = [x^{m+1}](x+1)^{m+n+1}$

the last x to choose is the i-th

$$\sum_{i=n+1}^{n+m+1} \binom{i-1}{n} = \sum_{k=0}^{m} \binom{n+k}{n}$$

Thm: $(x_1+x_2+\dots+x_m)^n = \sum_{n_1+n_2+\dots+n_m=n} \binom{n}{n_1, n_2, \dots, n_m} x_1^{n_1} \dots x_m^{n_m}$

k阶齐次线性递推关系

$\forall n \geq k$, $a_n = \sum_{i=1}^{k} c_i a_{n-i}$ (*), $c_1 \sim c_k$ are fixed constants ($c_k \neq 0$)

characteristic 方程: $x^k - \sum_{i=1}^{k} c_i x^{k-i} = 0$ (*')

q satisfies (*'), $\{q^n\}_{n\geq 0}$ satisfies (*)

Thm: (*') has k distinct roots $q_1 \sim q_k$, (*) has general solution

$$\left\{ \sum_{i=1}^{k} b_i q_i^n \right\} \quad b_1 \sim b_k \in \mathbb{R}$$

Pf: it obeys (*), to prove it is the solution

prove: $\forall a_1, \dots, a_n, \dots$ satisfying (*) can find $b_1 \dots b_k$ s.t $\sum_{i=1}^{k} b_i q_i^n = a_n$ ($\forall 0 \leq n \leq k-1$)

$$\begin{pmatrix} 1 & 1 & \cdots & 1 \\ q_1 & q_2 & \cdots & q_k \\ & & \vdots & \\ q_1^{k-1} & q_2^{k-1} & \cdots & q_k^{k-1} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_k \end{pmatrix} = \begin{pmatrix} a_0 \\ a_1 \\ \vdots \\ a_{k-1} \end{pmatrix}$$

since $q_1 \sim q_k$ are different, it's Vandermonde $\prod_{1 \le i < j \le k} (q_j - q_i) \neq 0$ ✓

<u>Thm</u>: (*) has roots $q_1, \cdots q_m$, q_i of multiplicity of r_i, $\sum_{i=1}^{m} r_i = k$

(*) has general solution: $\left\{ \sum_{i=1}^{m} P_i(n) q_i^n \right\}_{n \ge 0}$ $\deg P_i(n) \le r_i - 1$

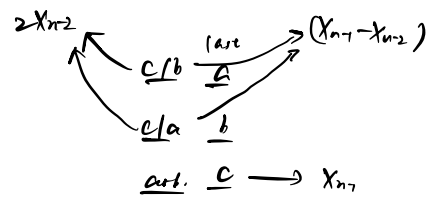
<u>Pf</u>: very difficult.

<u>Eg</u>: Find the # of n-sequences: letters = a.b.c ~~a~~ ~~b~~

<u>Sol</u>: Let it be x_n

$x_1 = 3$, $x_2 = 7$

$x_n = 2x_{n-1} + x_{n-2}$



<u>Eg</u>: (Hanoi's tower): $h_n = h_{n-1} + 1 + h_{n-1} = 2h_{n-1} + 1$

<u>Thm</u> k阶非齐次线性递推关系

$a_n = \sum_{i=1}^{k} c_i a_{n-i} + f(n)$ (*') general solutions $\{\xi_n + y_n\}_{n \ge 0}$ with

$\{\xi_n\}$ is one of the special solution of (*')

$\{y_n\}$ is the general solution of (*)

<u>生成函数</u>